

Car price prediction

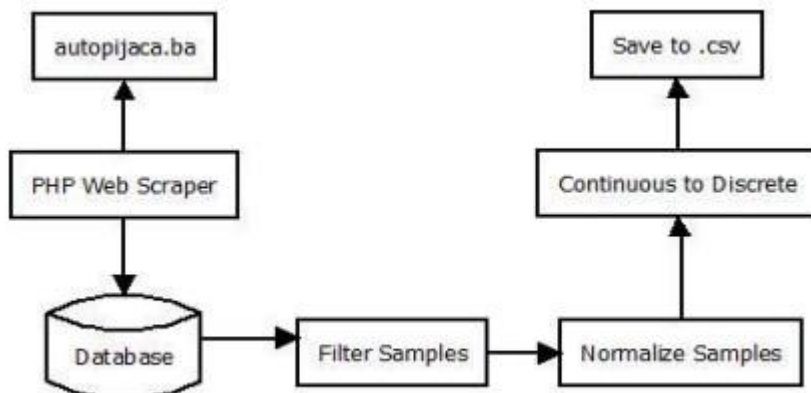
Literature survey by:

MEGHANA B-PES2UG20CS519

SUSHMITHA-PES2UG20CS472

[1] Enis Gegic, Becir Isakovic, Dino Keco, Zerina Masetic, Jasmin Kevric International Burch University, Sarajevo, Bosnia and Herzegovina. TEM Journal. Volume 8, Issue 1, Pages 113118, ISSN 2217-8309, DOI: 10.18421/TEM81-16, February 2019.

A car price prediction has been a high interest research area, as it requires noticeable effort and knowledge of the field expert. Considerable number of distinct attributes are examined for the reliable and accurate prediction. To build a model for predicting the price of used cars in Bosnia and Herzegovina, we applied three machine learning techniques (Artificial Neural Network, Support Vector Machine and Random Forest). However, the mentioned techniques were applied to work as an ensemble. The data used for the prediction was collected from the web portal autopijaca.ba using web scraper that was written in PHP programming language. Respective performances of different algorithms were then compared to find one that best suits the available data set. The final prediction model was integrated into Java application.



Materials and Methods: Data is collected from a local web portal for selling and buying cars autopijaca.ba [9], during winter season, as time interval itself has high impact on the price of the cars in Bosnia and Herzegovina. The following attributes were captured for each car: brand, model, car condition, fuel, year of manufacturing, power in kilowatts, transmission type, millage, colour, city, state, number of doors, four-wheel drive (yes/no), damaged (yes/no) etc. then the data collected in the web scraper is pre-processed and categorized with random forest like cheap, moderate, expensive finally classified with support vector machine and artificial neural networks **Model Implementation and**

Evaluation: Applying classification on cheap and moderate dataset using SVM and ANN algorithms.
Expensive dataset algorithms

Car price prediction can be a challenging task due to the high number of attributes that should be considered for the accurate prediction. The major step in the prediction process is collection and pre-processing of the data. In this research, PHP scripts were built to normalize, standardize and clean data to avoid unnecessary noise for machine learning algorithms.

[2] International Journal of Engineering and Advanced Technology (IJEAT)
ISSN: 2249 – 8958, Volume-9 Issue-1S3, December 2019 **Pattabiraman**
Venkatasubbu, Mukkesh Ganesh

The production of cars has been steadily increasing in the past decade, with over 70 million passenger cars being produced in the year 2016. This has given rise to the used car market, which on its own has become a booming industry. The recent advent of online portals has facilitated the need for both the customer and the seller to be better informed about the trends and patterns that determine the value of a used car in the market. Using Machine Learning Algorithms such as Lasso Regression, Multiple Regression and Regression trees, we will try to develop a statistical model which will be able to predict the price of a used car, based on previous consumer data and a given set of features.

Overfitting and underfitting come into picture when we create our statistical models. The models might be too biased to the training data and might not perform well on the test data set. This is called overfitting. Likewise, the models might not take into consideration all the variance present in the population and perform poorly on a test data set. This is called underfitting. A perfect balance needs to be achieved between these two, which leads to the concept of Bias-Variance trade-off. Pierre Geertz [2] has introduced and explained how biasvariance trade-off is achieved in both regression and classification. The selection of variables/attribute plays a vital role in influencing both the bias and variance of the statistical model. Robert Tibshirani [3] proposed a new method called Lasso, which minimizes the residual sum of squares. This returns a subset of attributes which need to be included in multiple regression to get the minimal error rate. Similarly, decision trees suffer from overfitting if they are not pruned/shrunk. Trevor Hastie and Daryl Pregibon [4] have explained the concept of pruning in their research paper. Moreover, hypothesis testing using ANOVA is needed to verify whether the different groups of errors really differ from each other. This is explained by TK Kim and Tae Kyun in their paper [5]. A Post-Hoc test needs to be performed along with ANOVA if the number of groups exceeds two

The prediction error rate of all the models was well under the accepted 5% of error. But, on further analysis, the mean error of the regression tree model was found to be more than the mean error rate of the multiple regression and lasso regression models. Even though for some seeds the regression tree has better accuracy, its error rates are higher for the rest. This has

been confirmed by performing an ANOVA. Also, the post-hoc test revealed that the error rates in multiple regression models and lasso regression models aren't significantly different from each other. To get even more accurate models, we can also choose more advanced machine learning algorithms such as random forests, an ensemble learning algorithm which creates multiple decision/regression trees, which brings down overfitting massively or Boosting, which tries to bias the overall model by weighing in the favour of good performers. More data from newer websites and different countries can also be scraped and this data can be used to retrain these models to check for reproducibility.

[3] e-ISSN: 2582-5208 International Research Journal of Modernization in Engineering Technology and Science Volume:03/Issue:03/March-2021 Impact Factor- 5.354

The world is growing day by day and also expectations of every people are also growing up. Out of all the expectation one of them is to buy a car. But all are not able to buy always a new car, so they will buy used one. But new person don't know about the market price for his or her dream car for old one. That is where we have need a platform which helps new people for car price prediction. In this paper we are coming up with that platform which is made using machine learning technology. Using supervised machine learning algorithms such as linear-regression, KNN, Random Forest, XG boost and Decision tree, let's try to build a statistical model which will be able to predict the price of a used car. For that previous consumer data and a given set of features will helps us. And we will also be comparing the prediction accuracy of these models to determine the optimal one.



Root Mean Squared Error, R-Squared Value KNN Regressor Plot Decision Tree Regressor Linear Regression Forest Regressor XG boost regressor are used for plotting purpose. they have used total five supervised machine learning models, and their root mean squared error are, KNN Regressor: 7771.09, Linear Regression: 6846.23, XG Boost: 3980.77, Random Forest: 3702.34 and Decision Tree Regressor: 5590.43. Out of all Random Forest has lowest RMSE, and performed well with highest Rsquared value: 0.93. The limitation of this research is less number of records of old car. in future if we get more data than we can retrain our models and might

get more accurate and stable model. This study used different models in order to predict used car prices. However, there was a relatively small dataset for making a strong inference because number of observations was only 92386. Gathering more data can yield more robust predictions. Secondly, there could be more features that can be good predictors. For example, here are some variables that might improve the model: number of doors, color, mechanical and cosmetic reconditioning time, used-to-new ratio and appraisal-to-trade ratio.

[4] Journal of Emerging Technologies and Innovative Research (JETIR)

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Mrs. T Veda Reddy, Y Praneeth, Y Sai Kiran, G Sai Pavan

It is easy for any company to price their new cars based on the manufacturing and marketing cost it involves. But when it comes to a used car it is quite difficult to define a price because it involves it is influenced by various parameters like car brand, manufactured year etc. The goal of this project is to create an efficient and effective model that will be able to predict the price of a used car by using the Gradient Boosting algorithm with better accuracy.

Data is collected from Kaggle.com [17]. The following attributes were captured for each car: Name, Location, Year, Fuel Type, Transmission, Owner Type, Mileage, Engine, Power, Seats, and Price expressed in Indian rupees. After the data was collected and stored, the data preprocessing step was applied. The attributes with unexpected values are processed accordingly i.e. In this case they have replaced them with the most repeated value of the attribute. The cars without a price are discarded in prior. To avoid conflict in mileage among different cars, all the mileages of cars are been scaled to a kmpl because most of the records are in km. To convert categorical data values into numeric attributes like Company, Name, Location, Fuel, Transmission, Owner, one-hot encoding [18] approach is used.

Gradient boosting technique is being used. It involves three elements: A loss function to be optimized, A weak learner to make predictions, an additive model to add weak learners to minimize the loss function. 70% of the data is used for training the model and 30% to test the model and evaluate its accuracy. Initially, 77% accuracy score was achieved by applying Gradient Boosting Regressor [20] to the data. After optimizing the parameters of the model by the Grid Search CV [21] method, the model has produced an accuracy score of 92%.

[5] Ijrasnet Journal for Research in Applied Science and Engineering

Technology, ISSN: 2321-9653, ESTD:2013, Paper Id : IJRASET44225, Publish Date : 202206-

13, Rithvik Raj Mekala, Gouru Laxmi Sevitha, Tushar Anumula, Kusuma Latha

Used car sales are on the rise all over the world. As a result, a method for estimating the value of a used car based on a variety of characteristics is urgently needed. Regression methods produce a continuous value rather than a categorized value as a put. As a result, instead of

estimating a car's price range, it will be possible to estimate the car's actual price. In addition, a user interface has been developed that takes user input and displays the pricing of an automobile based on that input.

Materials and Procedures Data is obtained from a local web portal for selling and buying cars, autopijaca.ba [9], during the winter season, the time interval has a significant impact on the pricing of cars in Bosnia and Herzegovina. Brand, model, automobile condition, fuel, year of manufacture, power in kilowatts, transmission type, mileage, color, city, state, and several doors were all recorded for each car. There are two basic phases: *Training Phase*: Using the data in the data set, the system is trained to fit a model (line/curve) depending on the algorithm selected. *Testing Phase*: The system is given inputs and its functionality is checked. The precision is tested. As a result, the data that is used to train or test the model must be appropriate. Because the system is intended to detect and estimate the price of a used car, distinct algorithms must be utilized for the two jobs. Different algorithms were compared for their accuracy before being picked for further use. The well-suited one for the task was chosen.

The proposed model consolidates two AI calculations: the Random Forest Method and the Extra Tree Regression strategy. This model loads the dataset at first so it very well might be explored further. This model was made with the assistance of a Kaggle Dataset. In the wake of performing information pre-handling steps on this dataset, for example, overseeing missing qualities and hot encoding of unmitigated factors, we start preparing the model for the circulated dataset into two 1. Preparing Dataset and 2. Test Dataset. This test information was picked aimlessly from the first dataset. For result expectation, the Random Forest Algorithm and the Extra Tree Regression Algorithm, as well as Randomized Search CV and Hyper-Parameters were used. When the model predicts an outcome, it will test it utilizing the test dataset made with the scikit-Learn module and measure its exactness. After experimenting we have come up with some results. The numeric values for each attribute in order will be taken as input to the Random Forest Regressor trained model and displays the output with the accuracy of 98 percent.

[6] International Journal of Innovative Research in Applied Sciences and Engineering (IJIRASE) Volume 4, Issue 2, DOI: 10.29027/IJIRASE.v4.i2.2020.629-632, August 2020, ISSN: 2456-8910

Predicting the price of used cars is one of the significant and interesting areas of analysis. As an increased demand in the second-hand car market, the business for both buyers and sellers has increased. For reliable and accurate prediction, it requires expert knowledge about the field because of the price of the cars dependent on many important factors. This paper proposed a supervised machine learning model using KNN (K Nearest Neighbour) regression algorithm to analyse the price of used cars. Model is trained with data of used cars which is collected from the Kaggle website. Through this experiment, the data was examined with different trained and test ratios. As a result, the accuracy of the proposed model is around 85% and is fitted as the optimized model.

The Used Cars data set was taken, and data processing has done to filter the data and to remove some unnecessary data. The model was trained with the processed data using KNN algorithm to predict the sales of used cars with higher accuracy. The data set is collected from a website Kaggle which is in a CSV format. The data set contains 14 variables which include an unnamed serial number, Name, Location, ileage, iFuel_Type, Engine a Transmission, Kilometers_Driven, Power, New_Price,Year, Seats, Owner_Type, Price.

Data Pre-processing: i) Removal of Non-numerical part from numerical features, ii) Converting Categorical values into numerical, iii) Separate the target variable. K Nearest neighbour algorithm is used to prepare a model which predict the price of used cars. It is a non-parametric method used for both regression and classification. It estimates the numerical target based on a similarity measure. A simple implementation of KNN is to find the average of the numerical target of the K nearest neighbours.

Cross-validation is used to inspect the overfitting of the model. It helps us to know about how predictive model can show the result for the new dataset. In this work, k fold crossvalidation is evaluated for the predictive model. In this, it splits the data set into k subparts and take one part as test data then the remaining part as the trained data. Then it will repeat the same steps for n times and produce n results. The average of the result has taken for the estimation value.

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